



T-34 tanks under construction in a Soviet factory

Great designers Mikhail Koshkin

As the head of the design team at the Kharkov tank factory in the Ukraine, Mikhail Koshkin's main legacy was the T-34, the medium tank that changed the course of World War II (see pp.98-101). Its background is rooted in the history of Soviet tank design.

UNDER JOSEPH STALIN'S leadership, the officials of the Soviet Union saw the tank as not only an important military asset, but also a vital symbol of power—and just like Adolf Hitler, the other major European dictator, Stalin took personal interest in the matter, influencing tank design and production in his country.

Russia, and later the Soviet Union, had a comparatively slow start in tank manufacturing. During World War I, no Russian tank designs saw production, but in the postwar period, captured vehicles such as the French Renault FT-17 were copied in Russian factories. As in other European countries, the 1920s and '30s was a period of experimentation in armored vehicle design, but Soviet industrial experience with heavy vehicles was scarce. Since it was a Communist power, the Soviet Union's only option



Mikhail Koshkin
(1898-1940)

for international industrial cooperation was with the other European pariah, Germany, which led to secret trials of German armored vehicles at a Soviet testing center at Kazan. As industrial experience and capacity grew with the Soviet Five Year Plans, new tanks were imported to copy and build under licence. These included the Vickers Mark E and Carden-Loyd tankettes from Britain, and an example of J. Walter Christie's M1931 wheel-cum-track vehicle

from the US. The latter was designed so that its tracks could be removed and it could run at speed on its wheels on roads. These tanks became the basis of major tank production in the Soviet Union, leading to the design of the T-26, BT-2, and T-27 vehicles.

Meanwhile, Mikhail Koshkin, who was born of humble stock, had been drafted into the army in 1917 and sent to various fronts. He later studied at a university and enrolled in technical college, ending up working in the Kirov factory in Leningrad on the T-29 and T-111 prototypes. By the time the Soviet military called for the development of a new replacement tank for the light BT series in 1937, Koshkin had risen to be the head of the design team at the Kharkov tank factory. He argued for abandoning wheel-cum-track vehicles, thickening vehicles' armor protection, and increasing the firepower of the proposed new models.

Despite internal arguments from rival factory teams and a lack of Red Army support, Koshkin presented his design directly to Stalin, who approved it. The vehicle became the famous T-34, a tank that began its service life with many mechanical and design issues—but successfully combined mobility, armor protection, and firepower. It was comparatively simple to manufacture and was produced in huge numbers, proving a fearsome surprise for the invading German Wermacht in 1941.



T-34 tank being assembled in Leningrad

The T-34 used J. Walter Christie's coil spring suspension system, but dispensed with the "convertible" track of earlier models, which could be removed for road use.

However, the T-34 was not the only Soviet tank in development at the time. A rival design team led by S. J. Kotin designed a new heavy tank—the KV (named after Kliment Voroshilov, the People's Commissar of Defense), which had heavy armor and the same 76 mm gun as the T-34. Like Koshkin, Kotin and his team argued that the earlier fashion for multiple turreted tanks had to be abandoned, and the KV had same diesel engine as proposed for the T-34 tank, lessening fire risks. Although the KV's variants were produced in much smaller numbers than the T-34, they subsequently formed the basis of other tanks including the heavy IS (Josef Stalin) series.

Perhaps the greatest achievement of Soviet tank manufacturing was the production of so many vehicles under such hardships. The German invasion not only led to the loss of huge numbers



Soviet T-34s on the move

The T-34's wide tracks and effective suspension gave it the ability to travel over muddy or snowy conditions.